MIL-L-3921A(AT)
5 January 1971
SUPERSEDING
MIL-L-3921
20 July 1954

MILITARY SPECIFICATION

LUBRICANT, MOLD, TIRE, SILICONE BASE EMULSION

1. SCOPE

1.1 Scope. This specification covers one type of silicone-base tire-mold lubricant, furnished as a concentrated emulsion (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Federal
O-E-760 - Ethyl Alcohol (Ethanol).
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

STANDARDS
Military
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials used in manufacture of the tire mold lubricant shall conform to requirements specified herein. Materials not specifically covered shall be of first commercial quality.

FSC-2640

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3.2 Composition. Tire mold lubricant emulsion shall consist of silicone liquid (dimethyl polysiloxane) of 200 to 500 centistoke viscosity at 77°F with such emulsifying, stabilizing, inhibiting, and other agents as are necessary to assure satisfaction of the requirements of this specification and satisfactory performance of the emulsion in intended use (see 6.1).

3.2.1 Silicone content. The silicone content, when determined as specified in 4.3.2 shall be not less than 34.0 percent by volume.

3.3 Color. Color of the tire mold lubricant shall, unless otherwise specified, be the "milk white" characteristic of water emulsions made from white or colorless ingredients (see 6.5). The color of the film shall not change objectionably at any temperature below 350°F.

3.4 Stability.

3.4.1 Undiluted emulsion. Tire mold lubricant emulsion shall show no creaming within 30 days storage and no phase separation within one year storage. When examined as specified in 4.2.2, and when tested as specified in 4.3.3.1 there shall be no phase separation (oil separation). Any creaming shall redisperse readily; gentle shaking or stirring shall restore a smooth, homogeneous mixture.

3.4.2 Diluted emulsion.

3.4.2.1 Soft water. When tested as specified in 4.3.3.2, the mold lubricant, diluted to a 70 to 1 ratio and to a 140 to 1 ratio with relatively "soft" water (see 4.3.1.2), shall show no more than slight creaming within 72 hours and no phase separation (oil separation) within two weeks. Creaming shall redisperse readily; gentle shaking or stirring shall restore a smooth, homogeneous emulsion.

3.4.2.2 Hard water. When diluted as specified in 3.4.2.1, except with relatively "hard" water (see 4.3.1.2), no phase separation (oil separation) shall occur within 24 hours and no more than a trace of creaming. Creaming shall redisperse readily, gentle shaking or stirring shall restore a smooth, homogeneous emulsion.

3.5 Temperature effects.

3.5.1 Low temperature. Repeated freezing and thawing in storage, or when tested as specified in 4.3.4.1, shall cause no phase separation (oil separation) and no creaming in excess of that permitted in 3.4.2.2.

3.5.2 High temperature. Repeated heating to 150°F and recooling to room temperature, in storage, or when tested as specified in 4.3.4.2, shall cause no separation and no creaming in excess of that permitted in 3.4.2.2

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3.6 Alkalinity. When tested as specified in 4.3.5, the ph (potential hydrogen content) of the emulsion at room temperature, or when heated to 200°F, shall be between 7.5 and 9.5. Numbers less than seven denotes increasing acidity. Numbers greater than seven denotes increasing alkalinity.

3.7 Odor. When diluted for mold application, the emulsion shall have no objectionable odor when applied to a hot mold.

3.8 Inertness. The tire mold lubricant, diluted or undiluted, shall not affect, or be affected by the metal shipping containers, mixing or spraying equipment. The emulsion shall show no carbonizing build-up, corrosion, or other unfavorable affect on smooth aluminum, ferrous alloy molds, or compounded rubber stocks when sprayed, brushed, or swabbed onto mold surfaces at any temperature below 350°F (see 6.3 and 6.4).

3.9 Instruction for use. The supplier shall furnish with each individual container, an approved label properly affixed, or appropriate printed instructions for diluted, application, and any precautions to assure proper handling and use (see 6.2).

3.9.1 Marking. Containers shall be labeled or marked "shake or stir well before using" (see 6.2).

3.10 Workmanship. Workmanship shall be such as to produce a product of the highest quality.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Sampling.

4.2.1.1 Lot formation. A lot or batch shall consist of silicone base emulsion, all of an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.2.1.2 Sampling for examination. Precautions shall be taken to assure that the sampling apparatus and specimens are neither contaminated with, nor altered by any material not representative of the lot being sampled. No less than five one-quart specimens shall be taken. If taken from a storage tank, specimens shall be taken from the top, bottom, and intermediate points by means of a sampling tube. If taken from a filling tank or discharge pipe, each specimen shall be taken at a different time during the filling operation. If taken from filled containers, the specimen containers shall be selected in such manner as to be representative of the lot. Specimens shall be taken and handled with extreme care to avoid unnecessary agitation prior to examination.

4.2.1.3 Sampling for tests. Upon completion of the examination procedure specified in 4.2.2, the specimens shall be combined and thoroughly mixed to form a uniformly homogeneous test sample from which the specimens for the required tests shall be taken.

4.2.2 Quality conformance examination. Each specimen handled as specified in 4.2.1.2 shall be examined for oil separation and for creaming, as evidence of instability. If any creaming is observed, the specimen shall be gently stirred or shaken and transferred to a transparent glass container for examination as to completeness of redispersion (see 3.4.1). Specimens shall also be examined for color and objectionable odor (see 3.3 and 3.7).

4.2.3 Quality conformance testing. Specimens selected in accordance with 4.2.1.3 shall be subjected to all tests specified herein, at the place of manufacture, under Government surveillance. Failure of any specimen to pass any specified test shall be cause for rejection of the lot.

4.3 Test procedures.

4.3.1 Test conditions.

4.3.1.1 Temperature. Except as otherwise specified, tests shall be conducted with fluids and emulsions at a stabilized temperature of 77 ± 7°F in still air at the same temperature.

4.3.1.2 Water hardness. For purposes of emulsion dilution, water designated as relatively soft (see 3.4.2.1) shall contain less than 50 parts per million (ppm) of solids. Water designated as relatively hard shall contain approximately 200 ppm of solids. Equal weights of calcium chloride ($CaCl_2$ anhydrous) and magnesium sulphate ($MgSO_4\ 7H_2O$) shall be added to distilled water to obtain the specified hardness.

4.3.2 Silicone content. To determine conformance to 3.2.1, a 10 milliliter (ml.) specimen of emulsion shall be taken from the test sample (see 4.2.1.3) and placed in a suitable 30 ml. centrifuge bottle so constructed and graduated that the volume of the fluid which separates can be read directly on the scale. A 20 ml. portion of ethyl alcohol conforming to grade I or II of O-E-760 or of butyl alcohol (98 percent or better) shall be added and well mixed with the

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emulsion by repeated inversion of the stoppered centrifuge bottle. Bottle shall then be placed in the centrifuge apparatus, properly counterbalanced, and the apparatus rotated for 10 minutes at 1,500 revolutions per minute (rpm). If a well defined interface does not result, bottle shall be shaken to redisperse contents and the process repeated. Since volume of specimen was 10 ml., the percent separated silicone fluid will be 10 times its volume in milliliters.

4.3.3 Stability.

4.3.3.1 Undiluted emulsion. To determine conformance to 3.4.1, a 30-ml. specimen from the test sample (see 4.2.1.3) shall be placed in a 30 ml. centrifuge tube, and 2 or 3 drops of water soluble dye (food coloring) added. The tube shall be inverted to disperse the dye and (with proper counterbalance) rotated for 30 minutes at 2700 to 3000 rpm after which careful inspection shall be made. (The dye concentration viewed by transmitted white light will indicate higher density in the "settled" position.) The centrifuge reading multiplied by 3.33 shall be recorded as the "stability".

4.3.3.2 Diluted emulsion. To determine conformance to 3.4.2.1 and 3.4.2.2, 1 ml. of the test sample (see 4.2.1.3) shall be transferred, with pipette, to 100 ml. or 250 ml. breakers, as required, and 70 ml. and 140 ml. of water of specified hardness, (see 4.3.1.2) shall be added, stirred well, covered with a watch glass, and at end of specified time shall be inspected for oil separation and creaming.

4.3.4 Temperature effects.

4.3.4.1 Low temperature. To determine conformance to 3.5.1, an undiluted 50 ml. specimen from the test sample (see 4.2.1.3) shall be placed in a suitable stoppered or covered container, and held for at least 1 hour in ambient air at a temperature of minus 40°F and then allowed to warm gradually in ambient air at 77° ± 7°F until stabilized at that temperature. Procedure shall be repeated through five thermal cycles, after which careful examination shall be made for separation or creaming. An immersion method, using dry ice and methanol as the cooling agent may be used in lieu of the air method.

4.3.4.2 High temperature. To determine conformance to 3.5.2, an undiluted 50 ml. specimen from the test sample (see 4.2.1.3) shall be placed in a suitable stoppered or covered container, held for 24 hours in a suitable oven at 150° ± 5°F, and then allowed to cool in still ambient air at 77 ± 7°F. Examination shall be made for discoloration and for any separation or creaming or other change in consistency or appearance.

4.3.5 Alkalinity. Following the high temperature test specified in 4.3.4.2, the same specimen shall be subjected to appropriate chemical tests utilizing suitable reagents and indicators. Tests shall be conducted with the specimen at room temperature and with specimen heated to 200°F to determine conformance to 3.6.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Unless otherwise specified (see 6.2), tire mold lubricant shall be packaged in metal cans with screw caps and inner seals conforming to PPP-C-96, type I. Tire mold lubricant shall be packaged in 1 pint, 1 quart, or 1 gallon capacity as specified (see 6.2).

5.2 Packing. Packing shall conform to PPP-C-96 for packing filled containers.

5.3 Marking. In addition to any special marking specified on the procurement documents, containers shall be marked to indicate whether packaging and packing is for domestic or oversea shipment, and for limited or extended storage.

6. NOTES

6.1 Intended use. Tire mold lubricant emulsion (mold release agent) is supplied as a concentrated emulsion for further dilution with 100 or more parts of water for spraying (or otherwise coating) tire molds, recapping or retread molds, tube plates, and the like, to form a thin, inert film to prevent adhesion of "rubber stocks" and to assure good surface finish of the cured compound.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and type of containers (see 5.1).
(c) Type of closure (see 5.1).
(d) Whether packing shall be for domestic or for oversea shipment (see 5.2).
(e) Any special instructions for labeling, packing, and marking (see 5.2 and 5.3).

6.3 Effects of water hardness. Mineral content of water used in making the highly dilute emulsion for spraying (or for other application) may influence creaming tendency, but deserves consideration from the more important effect on leaving deposits on polished surfaces to which it may be applied as a "release" agent (see 3.8). A few minute droplets, often called "fish-eyes", which occasionally leave a hard-water emulsion, if easy to redisperse are not to be classed as "phase separation".

6.4 Corrosive action. Experience has shown that the amount or corrosion inhibiting agent adequate to protect containers and equipment against any corrosive action by the undiluted tire mold lubricant emulsion (see 3.8) may at times be inadequate for completely affective protection of equipment from dilute emulsions made with water from some local sources. In this case the addition of proper amounts of sodium nitrite or other approved inhibitor has proved desirable.

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6.5 Color. Any color (or colored ingredient, see 3.3) which might cause the tire mold lubricant (release agent) to discolor rubber stocks (especially white) would be objectionable in commercial use but might be approved for military use if it imparts or contributes to any desirable performance characteristics of the tire mold lubricant (see 3.3).

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Preparing activity:
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Project No. 2640-0106

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
mil-L-3921A(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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